TABLE OF CONTENTS

TABLE OF CONTENTS 1
TABLE OF FIGURES 2
PRODUCTION BOARD TEST PLAN 3
HOW TO DONWLOAD NECESSARY TOOLS 3
REQUIRED SOFTWARE TOOLS 3
REQUIRED HARDWARE TOOLS 3
HOW TO INSTALL THE TOOLS? 4
HARDWARE INSTALLATION 4
HOW TO USE THE TEST TOOLS 6
HOW CAN WE MAKE A TEST? 6
SETTING UP THE HARDWARE 6
HOW TO INSTALL AND CONFIGURE TWINCAT 8

TABLE OF FIGURES

Figure 1 Headers to interconnect Control Board and Interface Board 4
Figure 2 Image showing power input headers 4
Figure 3 On-board LED showing power state 5
Figure 4 Programming header and connections 5
Figure 5 Device variant selection 6
Figure 6 Stepper motor connection headers 7
Figure 7 TwinCAT Start Page Menu View 8
Figure 8 Window showing compatible Ethernet Devices 8
Figure 9 Image showing driver successfully installed 9
Figure 10 Network settings properties option 9
Figure 11 Networking item view 9
Figure 12 TwinCat start page 10
Figure 13 TwinCat toolbar view 10
Figure 14 Solution explorer view 11
Figure 15 I/O device view 11
Figure 16 Scan for boxes pop-up 11
Figure 17 Device tab section view 12
Figure 18 Device properties tab 12
Figure 19 Device settings list 13
Figure 20 EEPROM update section 13
Figure 21 Twincat project view 14
Figure 22 Device tab section view 14
Figure 23 Twincat project view 15
Figure 24 Image showing option from where TwinCAT can be started in Run Mode 15
Figure 25 BIOS Menu in an HP System 16
Figure 26 Virtualization Technology (VTx) checkbox image 16
Figure 27 Screenshot showing location of Activate Configuration 17
Figure 28 Generate License Prompt Box 17
Figure 29 Screenshot showing login screen 17
Figure 30 Start button in TwinCAT 18
Figure 31Main control window for testing firmware 18

PRODUCTION BOARD TEST PLAN

HOW TO DONWLOAD NECESSARY TOOLS

This section will cover the basic requirements from both software and hardware perspective which will allow us to power up the board, load the firmware and then test it.

REQUIRED SOFTWARE TOOLS

1. J-Link Commander (Select 32-bit Installer or 64-bit Installer depending on your OS)
2. J-Link serial port drivers (Although these are installed automatically upon plugging in, if not, please do a manual install)
3. TwinCAT (For sending EtherCAT commands)

REQUIRED HARDWARE TOOLS

1. Interface Board and Control Board PCBs
2. Power Supply 24V, 3A
3. J-Link Probe
4. J-Link probe converter (will be provided by eSpark for easy firmware upload)
5. Stepper Motor

HOW TO INSTALL THE TOOLS?

HARDWARE INSTALLATION

To install the hardware and make it ready for firmware upload, following steps are required:

1. Connect Interface Board with Control Board using interconnect jumpers provided on both boards as shown below

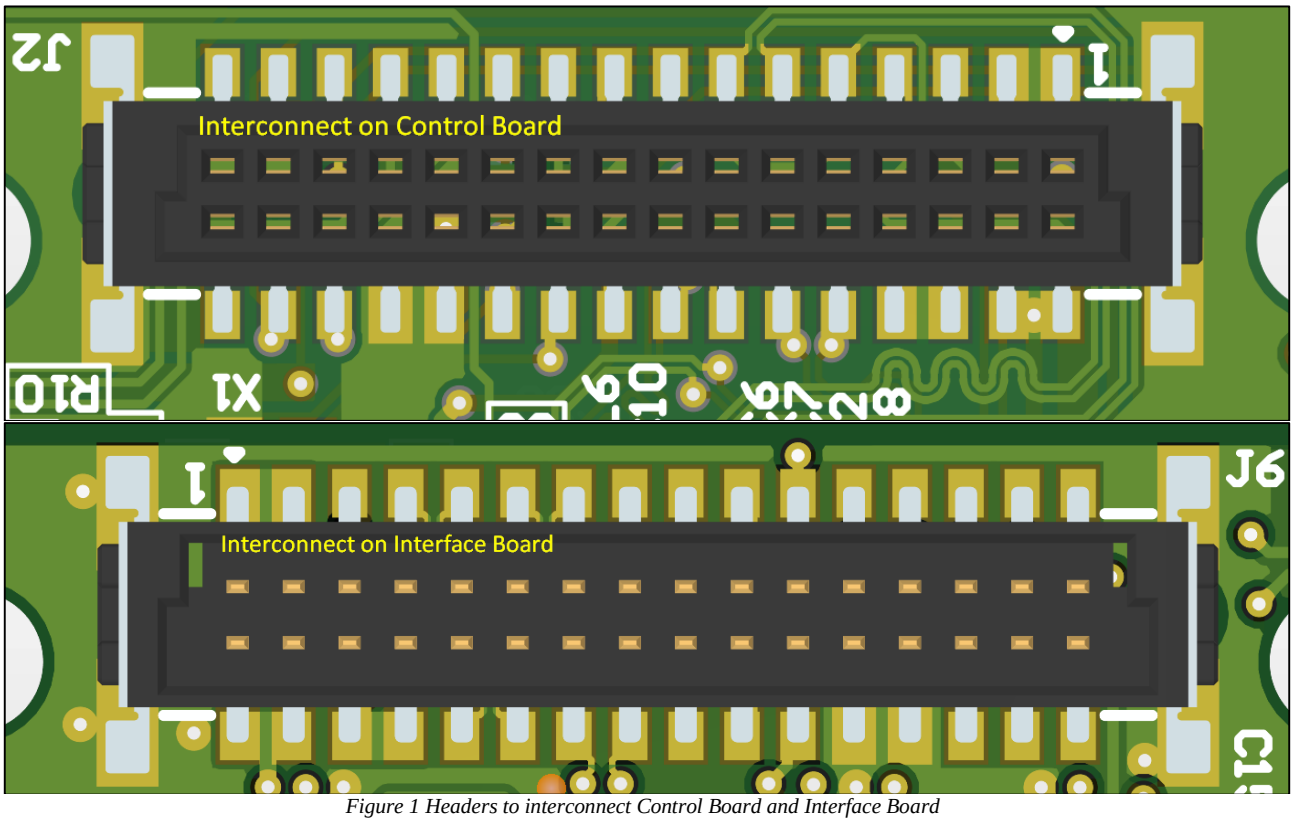


Figure 1 Headers to interconnect Control Board and Interface Board

2. Power up the board using adequate power supply mentioned in specifications above. The power input is to be applied on the headers shown in image below:

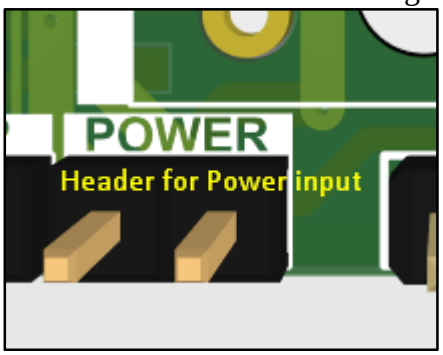


Figure 2 Image showing power input headers

3. Once the board is successfully powered up, an on-board test LED on the **Control Board** will light up indicating successful power up as shown below



Figure 3 On-board LED showing power state

4. Connect the provided J-link probe to the programming header as shown in the image below:

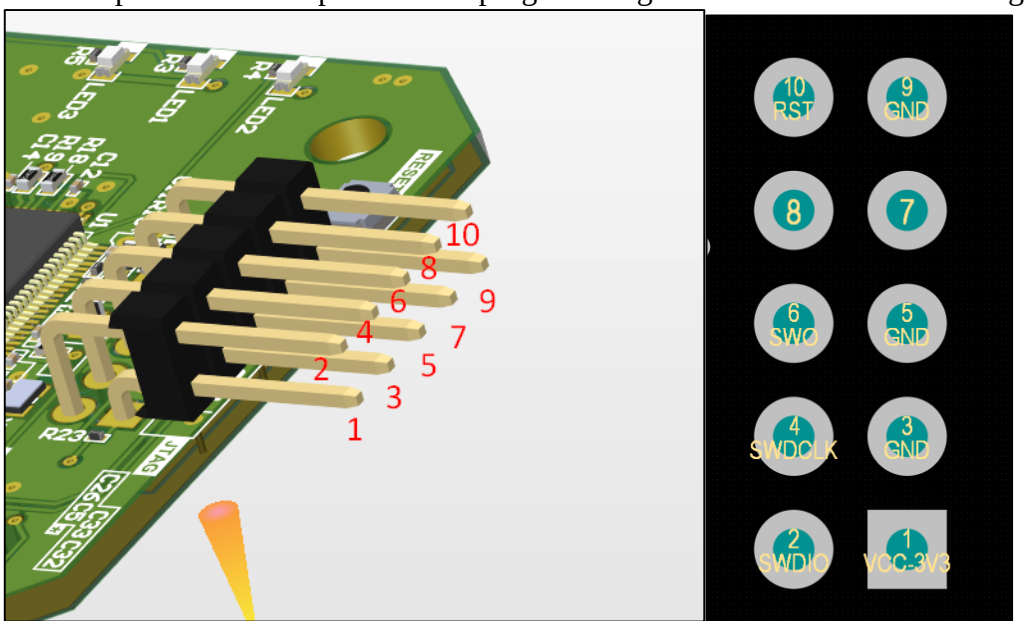


Figure 4 Programming header and connections

5. Once the programming probe is connected, launch **J-Link Commander** and run the **connect** command and follow the method shown below.



6. In new opened window, select the device variant by typing "ATSAME53J19A" or selecting the appropriate variant.

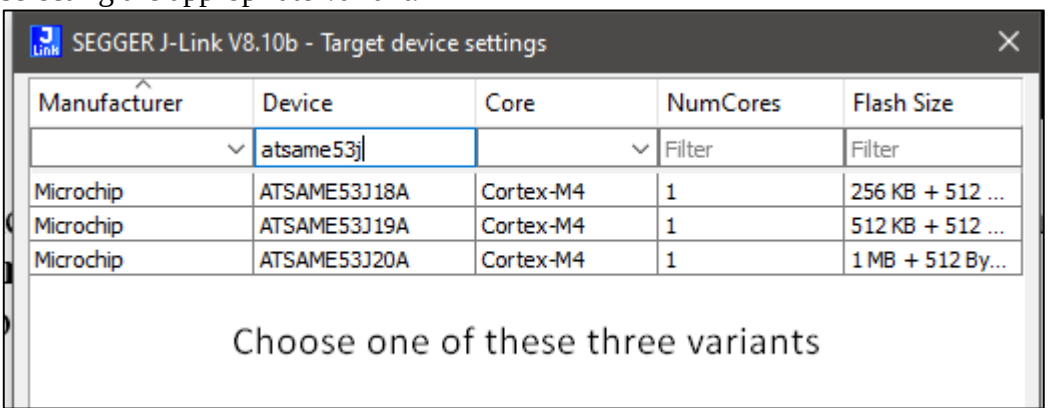


Figure 5 Device variant selection

7. Select "SWD" by typing "s" as target interface and select default speed.
8. If the connection is successful, this means the processor is alive and working well.

HOW TO USE THE TEST TOOLS

1. After successfully connecting with the hardware using J-Link Commander, upload the provided **Final_Test.elf** file which is the final compiled code file for the project. To do this, run the command:

```
LoadFile <FileName> [<Addr(.bin only)>] [<noreset | reset>]
      loadfile <path\to\provided\elf\file> (Example)
```

2. After successful upload, the J-Link Commander will show **successfully loaded**.

HOW CAN WE MAKE A TEST?

SETTING UP THE HARDWARE

In order to prepare a test, follow the steps below:

1. Connect the stepper motor to the header labeled **MOTOR OUT**. Connections are also enlisted starting from left side:

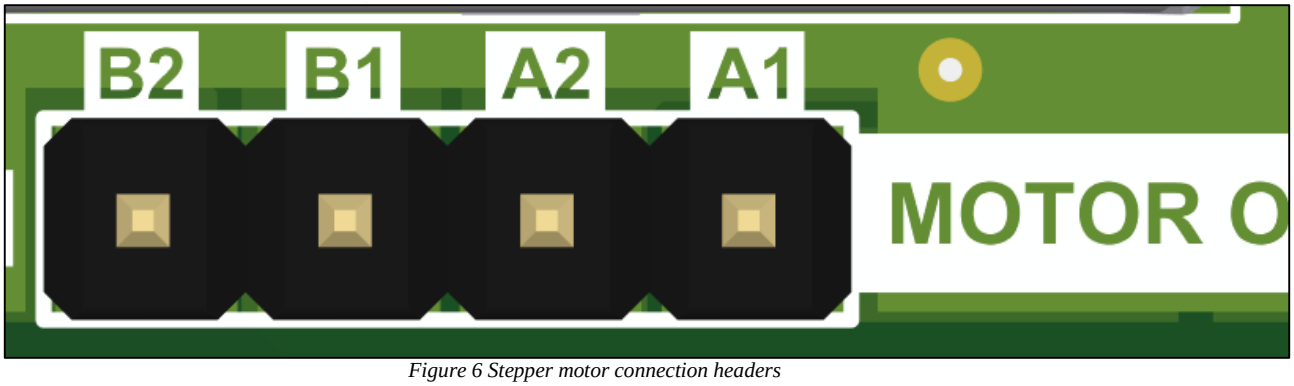


Figure 6 Stepper motor connection headers

a. **MOTOR_A1 (Motor coil A1)**
b. **MOTOR_A2 (Motor coil A2)**
c. **MOTOR_B1 (Motor coil B1)**
d. **MOTOR_B2 (Motor coil B2)**

2. Once the connections are done, power up the board and connect the Ethernet cable from **PC** to the board.
3. Observe the commands behavior of board on the **TwinCAT** software

HOW TO INSTALL AND CONFIGURE TWINCAT

1. Run the installation file.
2. The on-screen menu will guide with the installation process. Please note that we have to select **Complete Installation** option.
3. Once the installation is complete, launch TwinCAT software and go to **TwinCAT > Show Realtime Ethernet Compatible Devices…**

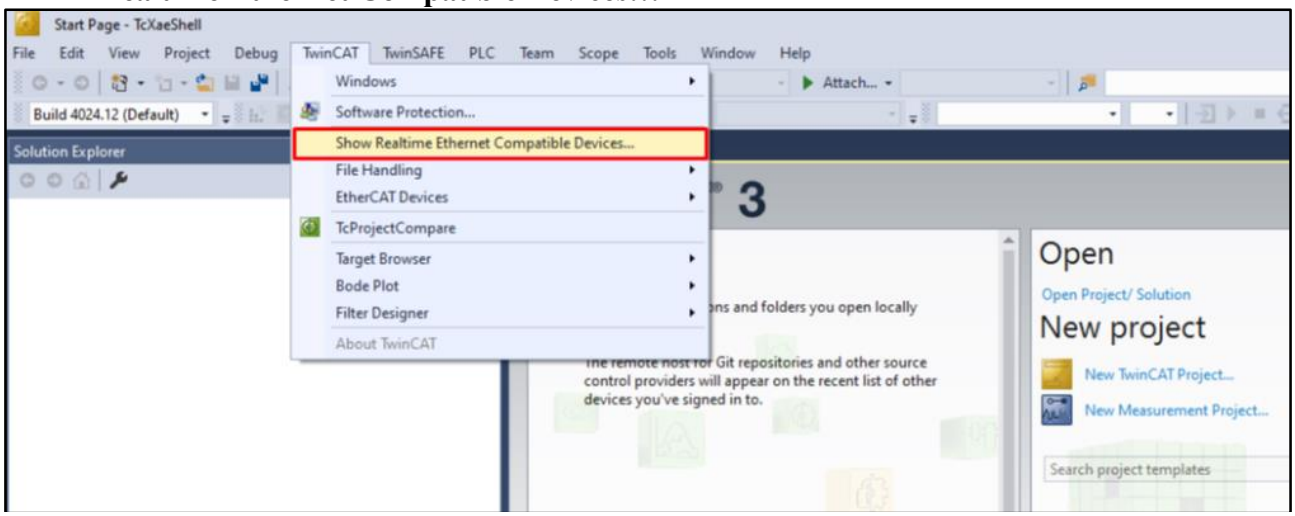


Figure 7 TwinCAT Start Page Menu View

4. The following window will open. Select the driver as shown and click **Install**

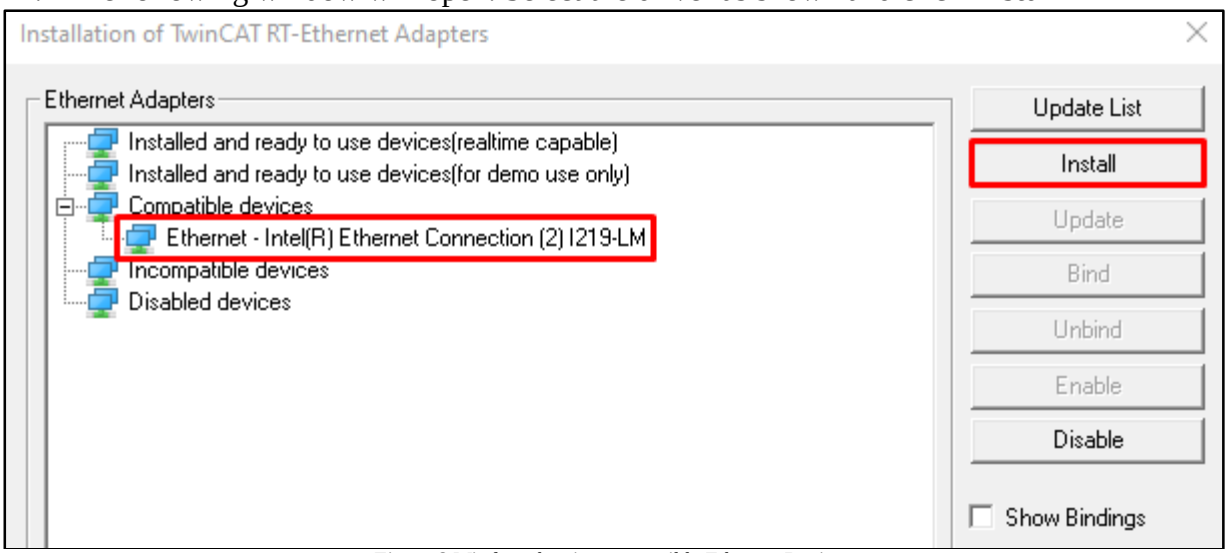


Figure 8 Window showing compatible Ethernet Devices

5. Once the driver is successfully installed, it will appear under installed devices as shown below:

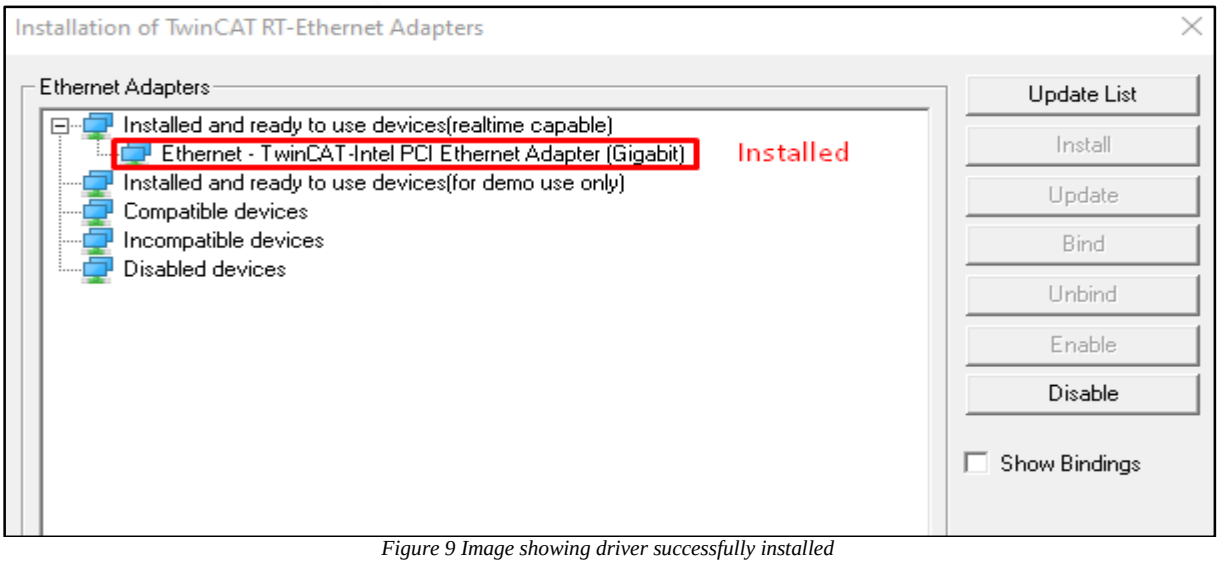


Figure 9 Image showing driver successfully installed

6. After verifying in TwinCAT, go to **Network Settings** in computer and open **Properties**

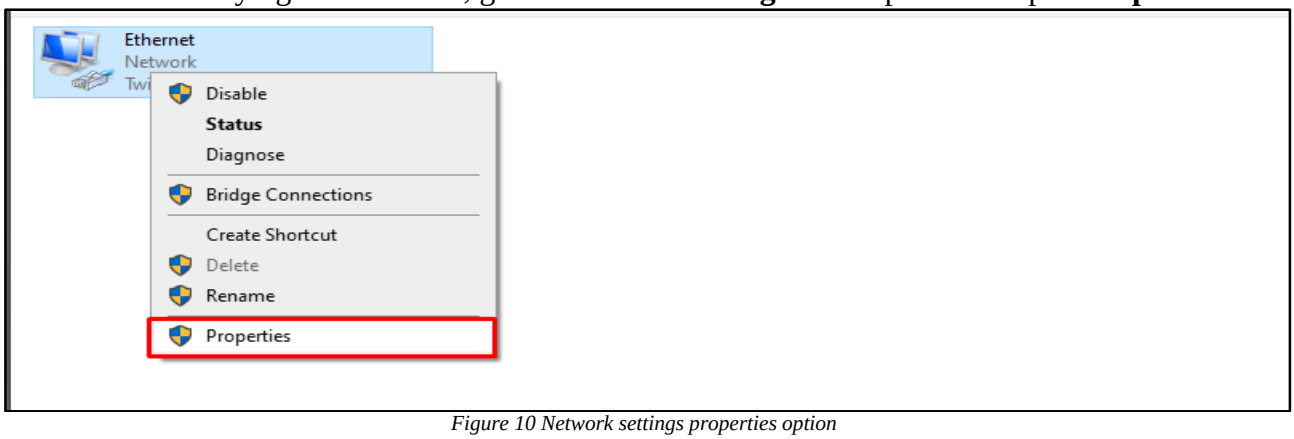


Figure 10 Network settings properties option

7. Disable all the items and only allow **TwinCAT Ethernet Protocol** to remain enabled

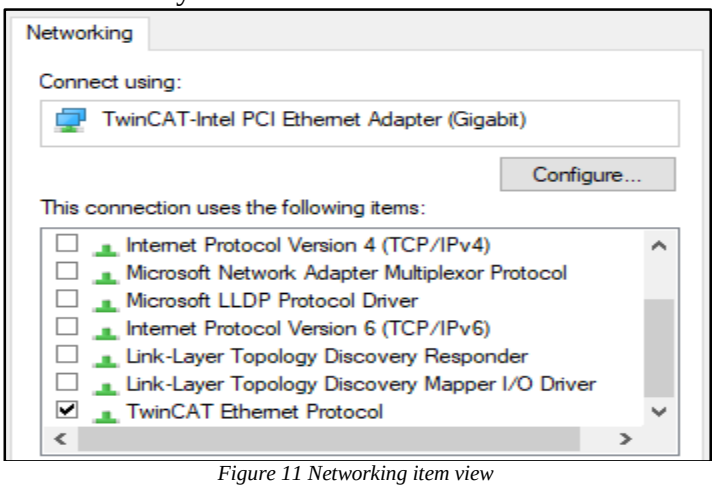


Figure 11 Networking item view

8. Place the LAN9255.xml file in the following path C:\TwinCAT\3.1\Config\Io\EtherCAT

9. Create a new Project by clicking on **New TwinCat Project.**

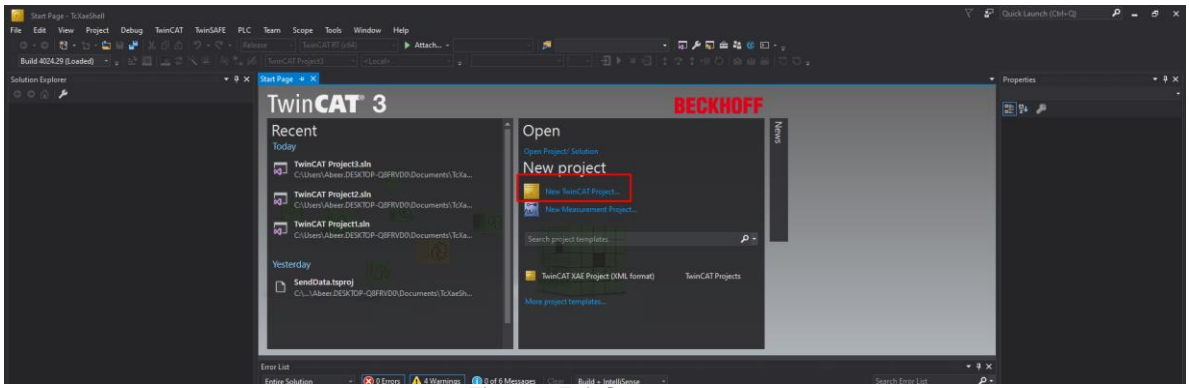


Figure 12 TwinCat start page

10. Switch to config mode.

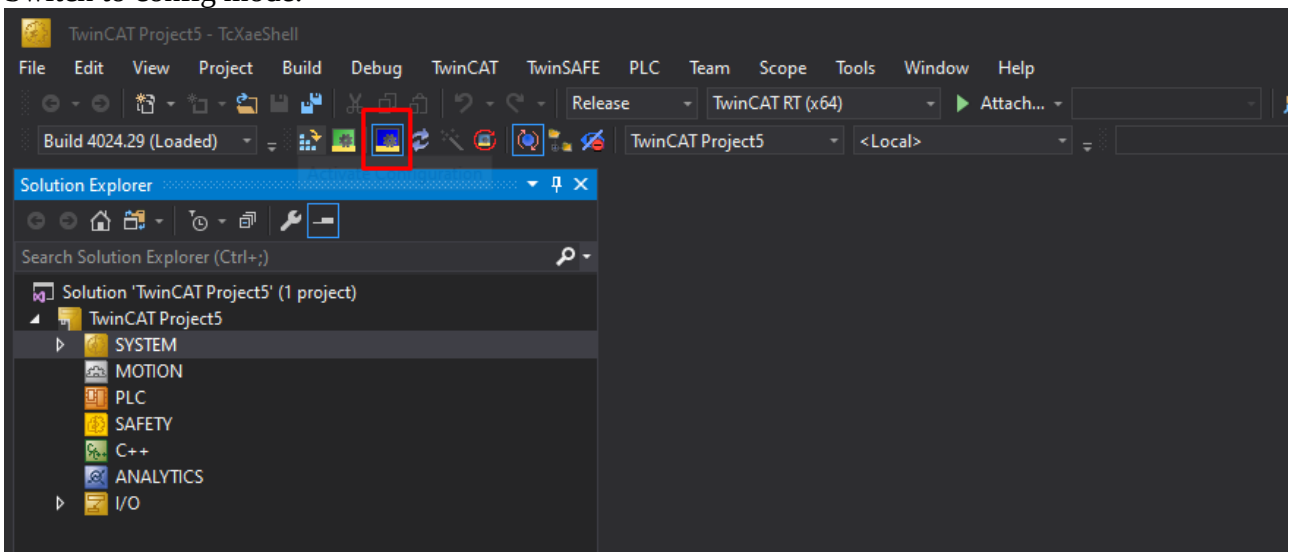


Figure 13 TwinCat toolbar view

11. Make sure that the ethernet cable is connected to the **Port A** of the board then click on Scan.

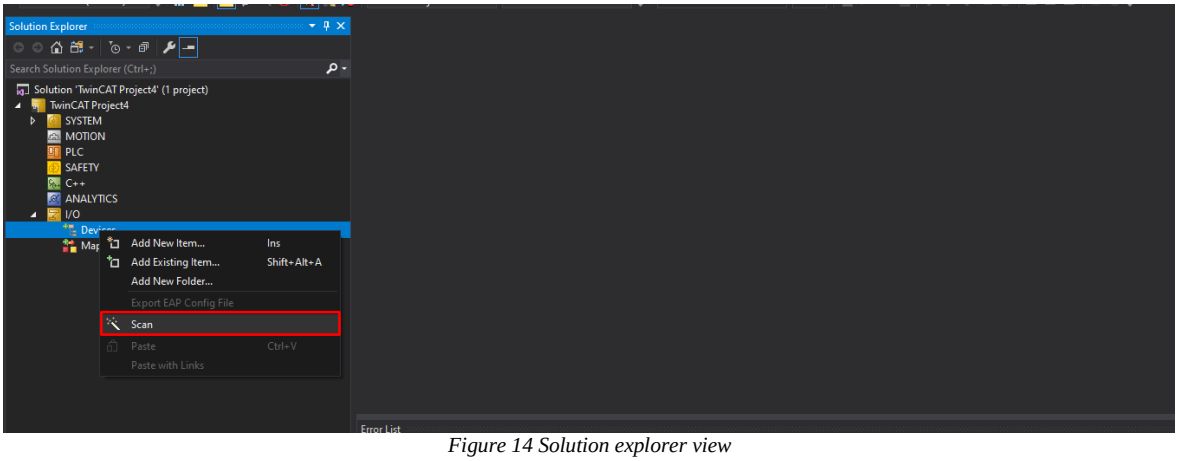


Figure 14 Solution explorer view

12. The board will appear as Device 2. Click on OK.

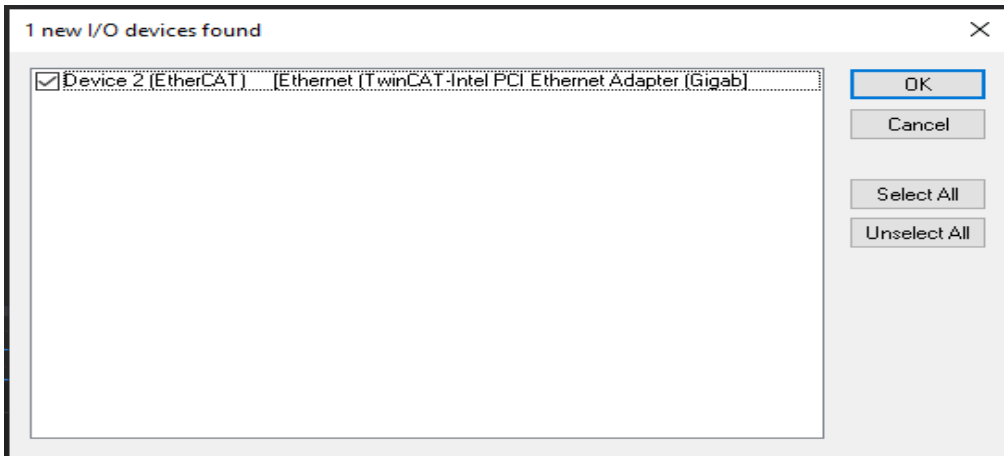


Figure 15 I/O device view

13. Click Yes.

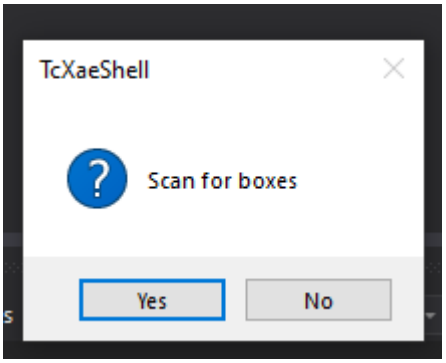


Figure 16 Scan for boxes pop-up

14. Since the EEPROM is not there So the board will appear like this.

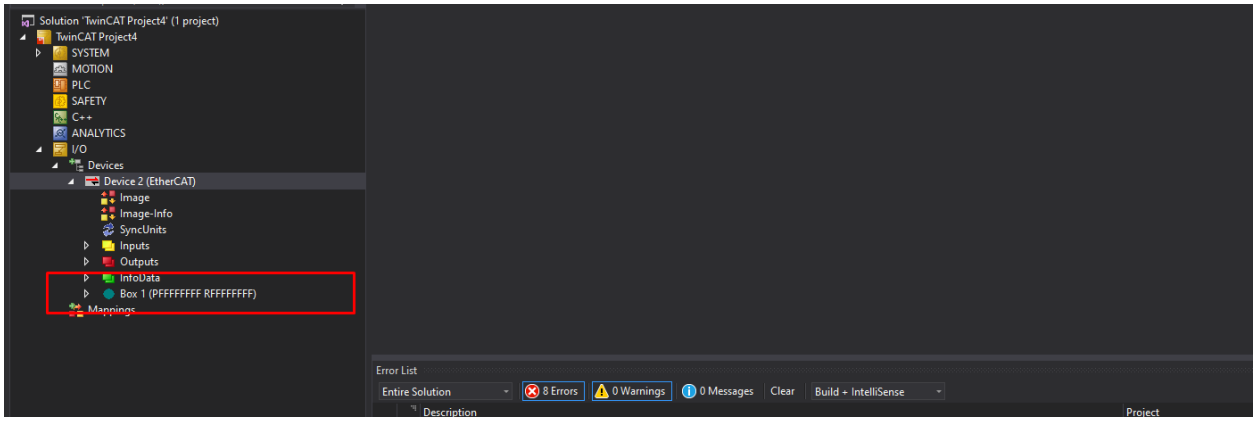


Figure 17 Device tab section view

15. Navigate to **Device 2** and the to the **Online** tab.

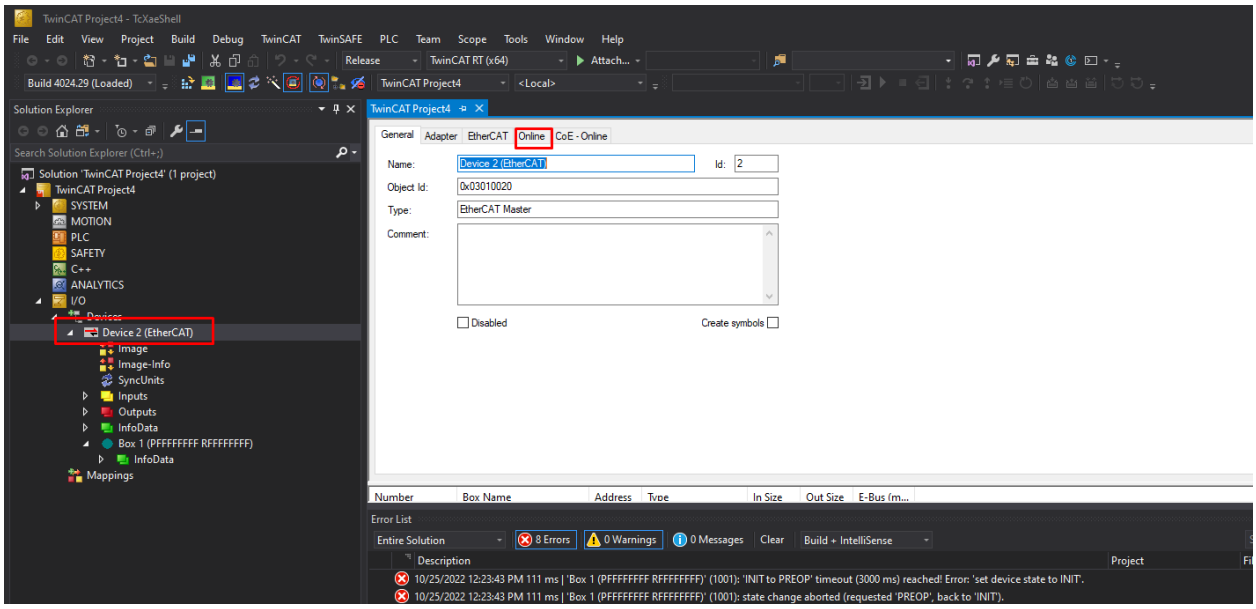


Figure 18 Device properties tab

16. Right click on Box 1 and then select **EEPROM Update**.

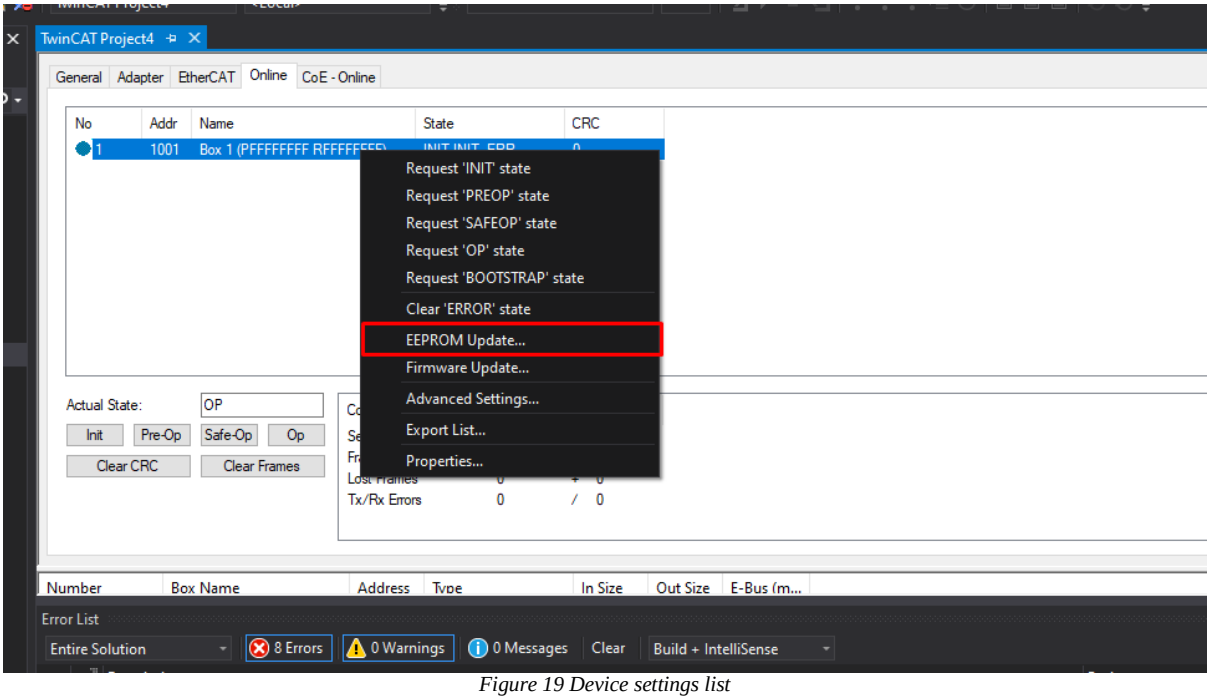


Figure 19 Device settings list

17. Scroll down till you find Kurvatur A/S and then double click on sample_app to flash the EEPROM.

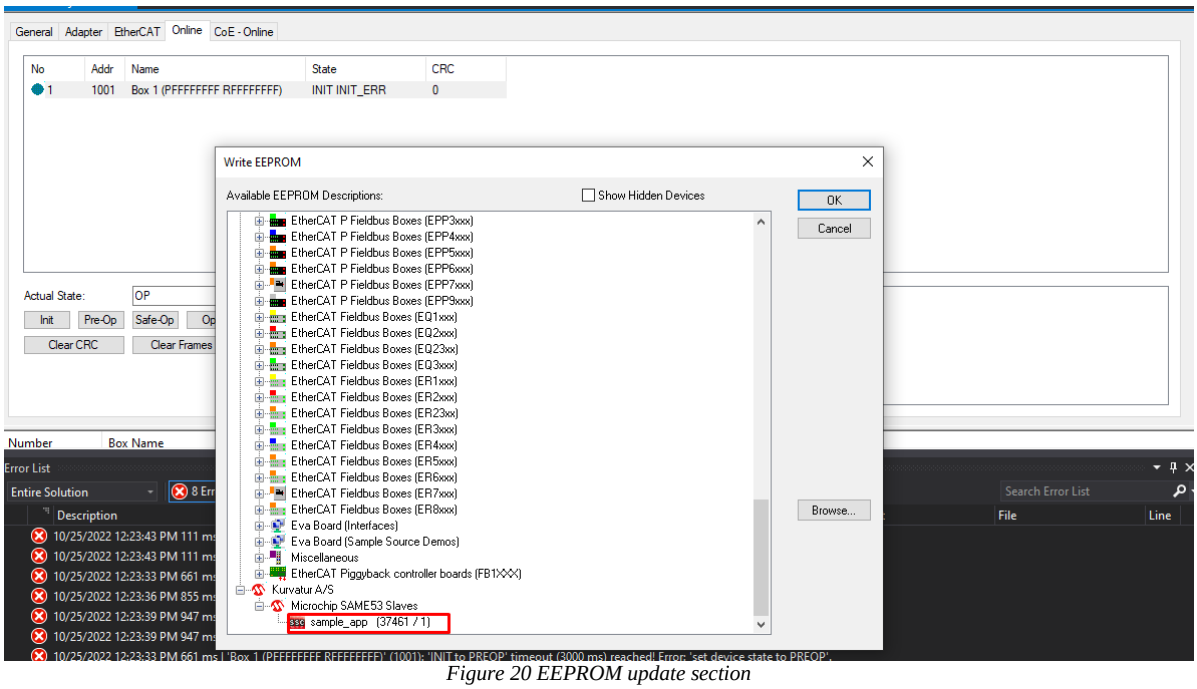


Figure 20 EEPROM update section

18. The EEPROM is now flashing.

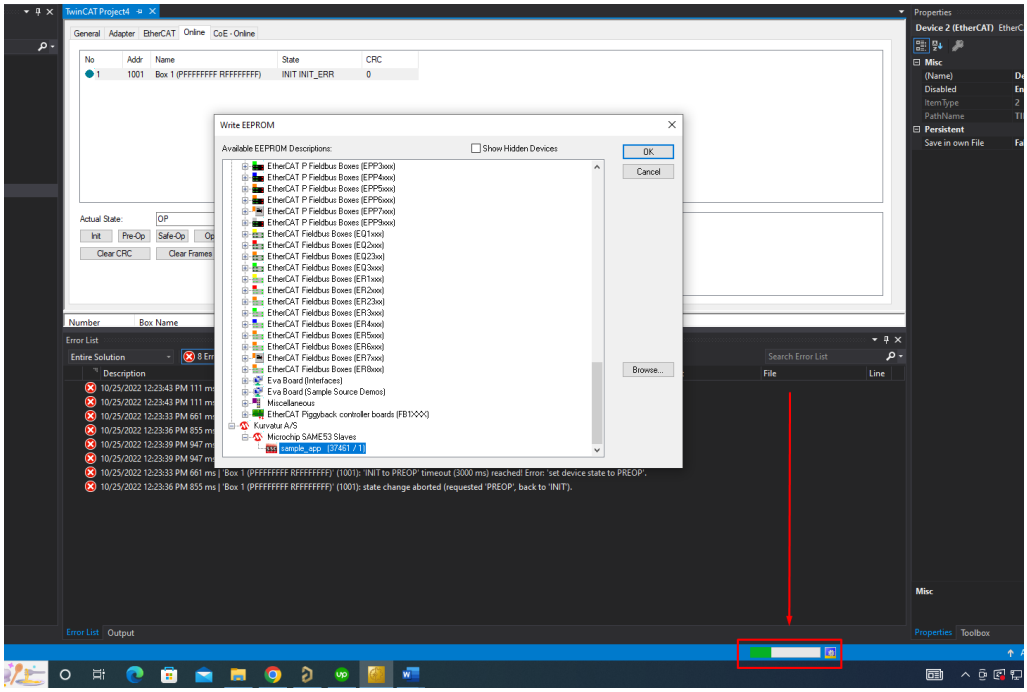


Figure 21 Twincat project view

19. Once the flashing is done remove the device and then scan again.

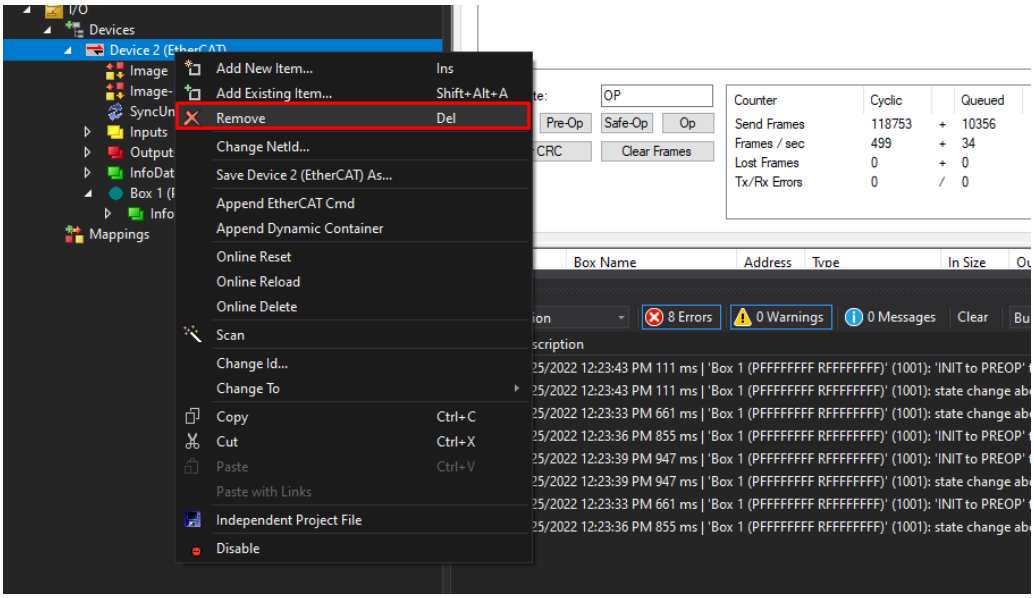


Figure 22 Device tab section view

20. Box 1 will be displayed as sample_app now. EEPROM is flashed successfully.

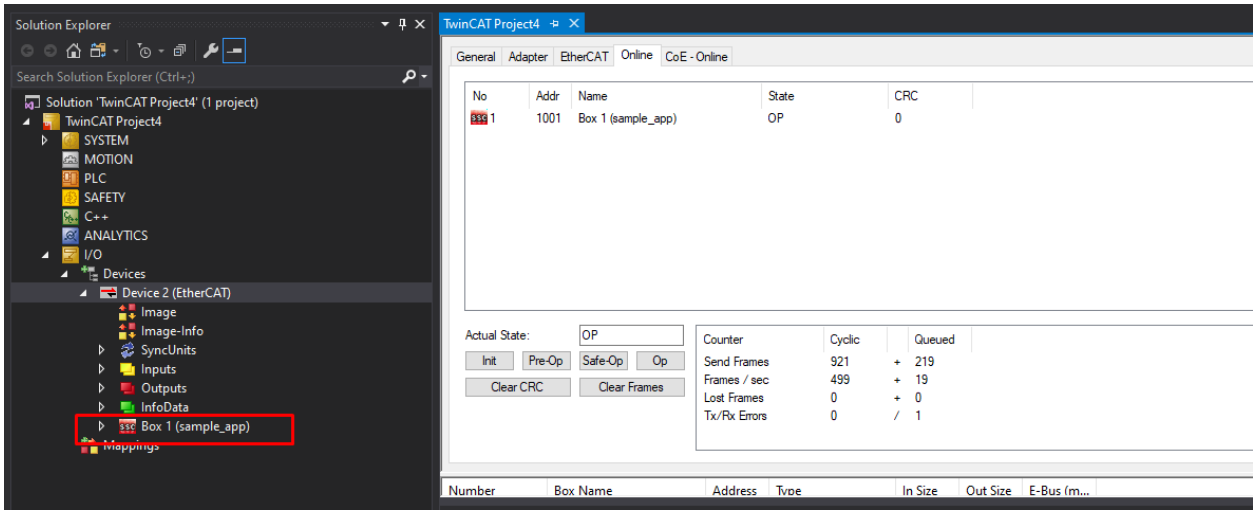


Figure 23 Twincat project view

21. Open the provided **SendData TwinCAT Project**. The project file is located in the main folder as **SendData.sln**. Please make sure the project is opened using **TwinCAT XAE Shell Software only**.
22. Once the project is opened, click the **Restart TwinCAT System** button as shown in the image below. This will restart TwinCAT in **Run Mode**.

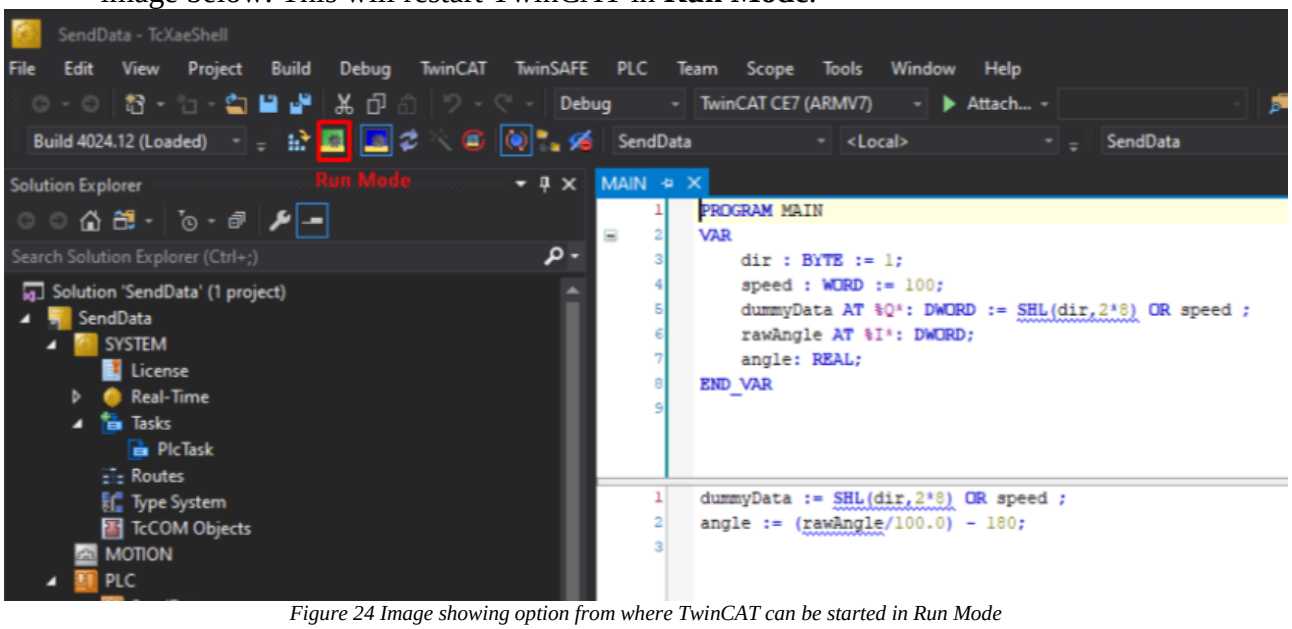


Figure 24 Image showing option from where TwinCAT can be started in Run Mode

23. If the system responds with following two errors, this means vt-x is disabled and has to be enabled from BIOS.
24. The setup assumes an **HP System** for explaining how to enter and enable vt-x (virtualization) from **BIOS**. When the computer restarts, press **ESCAPE** key to enter **BIOS Menu**.

25. Once inside the menu, the following options will appear:

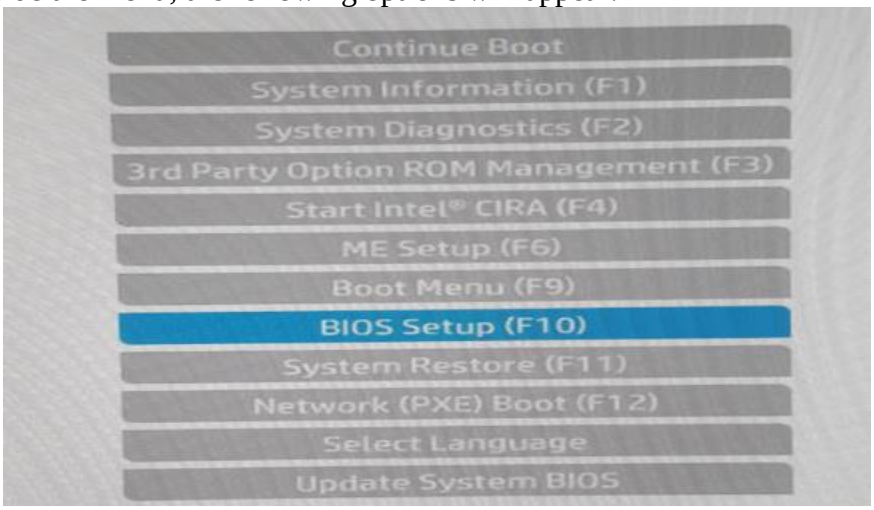


Figure 25 BIOS Menu in an HP System

26. Go to **Advanced > System Options** and enable **Virtualization Technology (VTx)**.

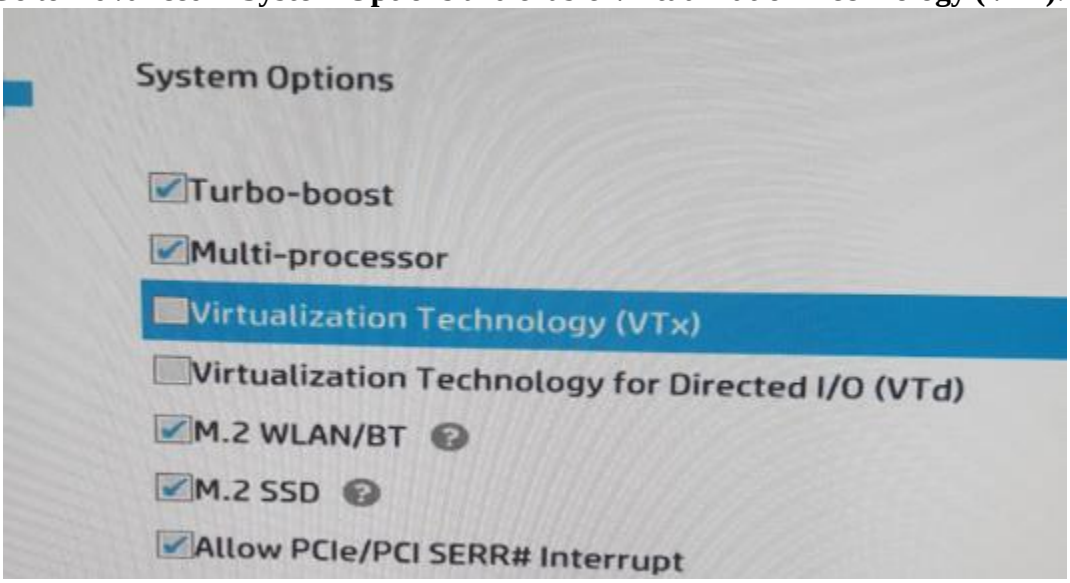


Figure 26 Virtualization Technology (VTx) checkbox image

27. After this, when windows opens, Go to **C:\TwinCAT\3.1\System** and find **win8settick.bat**. Run is as **Administrator** and reboot the machine.

28. When PC restarts again, launch **TwinCAT XAE Shell** and click on **Activate Configuration** as shown in the image below:

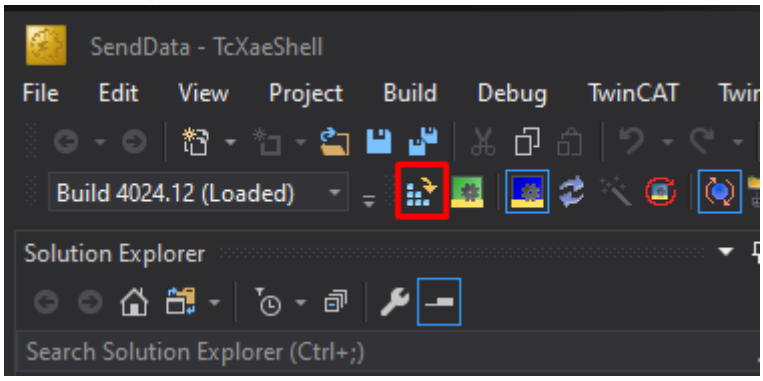


Figure 27 Screenshot showing location of Activate Configuration

29. A prompt will appear asking for generation of licenses. Click **Yes** to proceed.

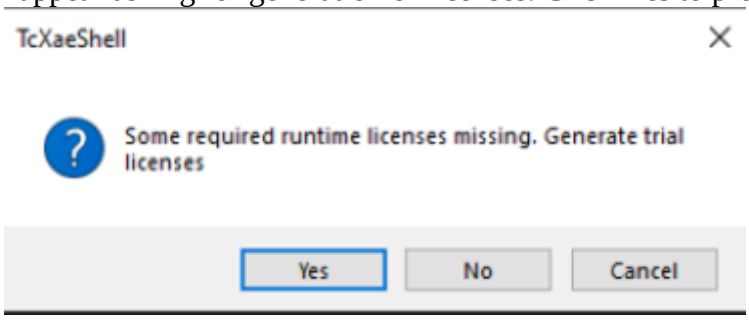


Figure 28 Generate License Prompt Box

30. Enter **Security Code** and press **O.K.** to proceed. After this, place the shared configuration file **LAN9255.xml** at following location **C:\TwinCAT\3.1\Config\Io\EtherCAT**
31. After file placement, restart **TwinCAT XAE Shell** and repeat **Step 9**. After TwinCAT restarts, click **Login** as shown:

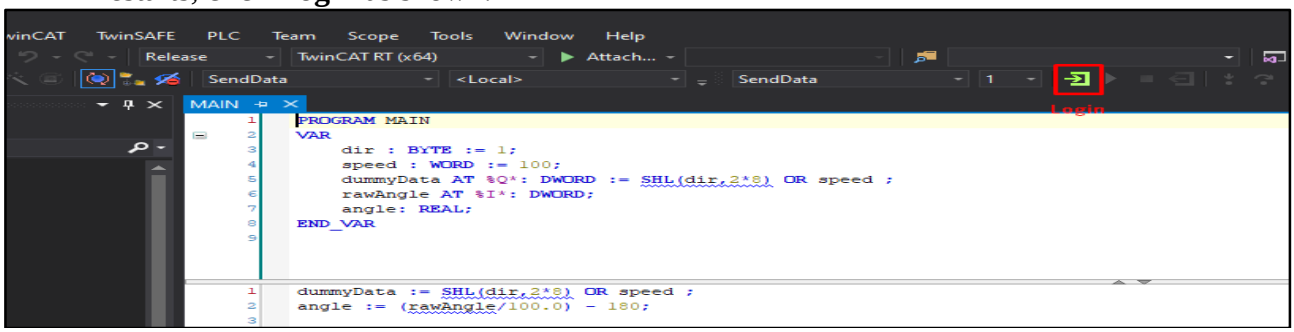


Figure 29 Screenshot showing login screen

32. After that, click **Start** button.

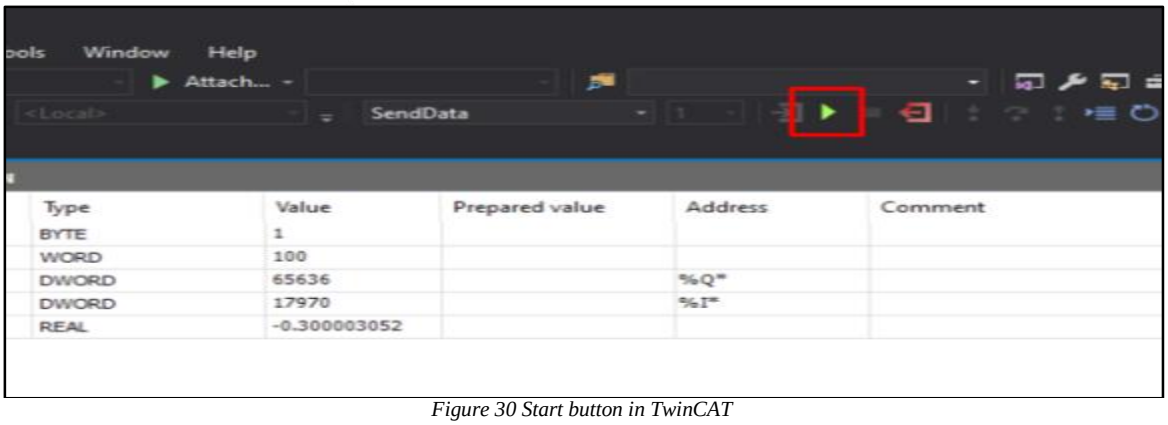


Figure 30 Start button in TwinCAT

33. Now you can control speed and direction by preparing it for another value and then write the values and simultaneously receive motor position in the angle variable

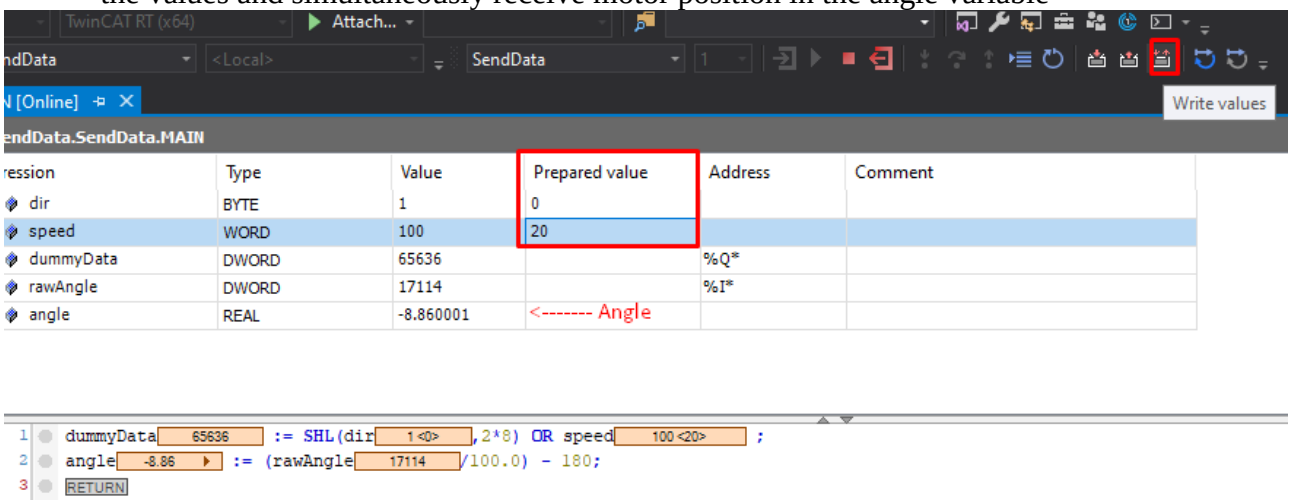


Figure 31Main control window for testing firmware